ECS 171 MID-PROJECT REPORT: Does Naive Bayes Really Learn in $O(\log n)$?

Measuring the Generative-Discriminative Crossover as Dimensionality Grows

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https://github.com/ango-dev/ECS171-FQ26-Group24

1. Assigned Paper and Project Motivation

This project tests the claims in Ng and Jordan (2001), *"On Discriminative vs. Generative Classifiers: A Comparison of Logistic Regression and Naive Bayes."* The paper compares a generative classifier (naive Bayes) against a discriminative classifier (logistic regression) under a shared feature assumption. It posits that naive Bayes requires $O(\log n)$ training examples to approach its asymptotic error, while logistic regression requires $O(n)$ examples but achieves a lower final error. Consequently, for a fixed number of features n, there is a theoretical crossover point $m^*(n)$ in training-set size where logistic regression outperforms naive Bayes. Spam classification provides a genuine binary text-classification task to empirically test these variables (m, n, and test error) using real text data rather than simulated datasets.

The authors support the result with learning curves on 15 small UCI datasets but never vary n directly, so the $\log n$ versus n rates themselves are untested. With bag-of-words text the vocabulary size can be set to any value, which turns the rate claim into a measurable prediction about how the crossover moves as n grows.

2. Problem Statement and Research Objectives

Does naive Bayes learn at the predicted $O(\log n)$ rate relative to logistic regression on real datasets? Our primary objectives are:

- Measure test-error learning curves for both models as a function of training-set size (m) at fixed vocabulary sizes (n).
- Identify the crossover point $m^*(n)$ where logistic regression achieves a lower test error than naive Bayes for each n.
- Determine if $m^*(n)$ scales with n according to the paper's asymptotics using log, linear, and power-law fits.
- Assess whether this relationship holds across corpora of varying sizes and class balances.
- Check whether the early naive Bayes advantage also appears under log-loss, or only under 0–1 misclassification error.

Currently, exploratory data analysis (EDA) and preprocessing are complete, but model training and objective testing have not started.

3. Dataset Description

The primary dataset is the SMS Spam Collection, which contains 5,159 English messages (~87% ham / 13% spam) after deduplication. Because the usable training-set ceiling ($m_{\max}$) at $n = 1{,}024$ is only $3.1 \times n$, the vocabulary grid is capped at $n = 2{,}048$. If the SMS data yields inconclusive results at larger n, the Enron-Spam corpus will serve as a larger, balanced contingency dataset. Features are represented as a binary bag-of-words using a custom token pattern (`[a-z0-9]+|[£$€!]`)

that retains high-signal digits and currency symbols. All data splits are class-stratified to prevent single-class samples. The table below summarizes both corpora as measured in the EDA; $m_{\max}$ is the experiment pool minus the test set.

	SMS Spam Collection	**Enron-Spam**
Messages as downloaded (spam share)	5,574 (13.4%)	33,716 (50.9%)
Messages after deduplication (spam share)	5,159 (12.4%)	30,138 (48.3%)
20% dev split: messages / distinct tokens	1,031 / 3,362	6,028 / 65,469
Largest training set $m_{\max}$	3,128	22,110
Vocabulary grid n	16 to 2,048	16 to 4,096

4. Exploratory Data Analysis (EDA)

EDA on both raw corpora informed key choices: (1) *Exact duplicates:* Deduplicated to prevent artificial accuracy inflation across train/test splits. (2) *Class balance:* Stratified splits guarantee proper class representation in skewed datasets. (3) *Surface cues:* Digits and symbols appear in ~77% of spam vs. ~0.1% of ham, prompting custom tokenization. (4) *Vocabulary growth:* ~56% of tokens occur once; vocabulary is ranked by mutual information on a held-out split. (5) *Feasibility:* The ceiling on SMS training size necessitates the Enron-Spam contingency plan for larger m-sweeps.

Exact duplicates. SMS contains 415 extra copies (7.4% of messages) and Enron 3,578 (10.6%). Spam is duplicated about twice as often as ham in both corpora (14.1% vs. 6.4% in SMS; 15.3% vs. 5.8% in Enron). No duplicated text carries two different labels, so copies can be dropped safely.

Class balance. After deduplication SMS is 12.4% spam and Enron 48.3%. The imbalance matters most at small m: a stratified training set of $m = 64$ holds about 8 spam messages, and an unstratified set of $m = 8$ has roughly a one-in-three chance of containing no spam at all.

Message length and surface cues. SMS spam is longer than ham (median 28 vs. 11 tokens). The ~77% figure is for numbers of five or more digits; "£" appears in 33% of spam and "!" in 47%. scikit-learn's default token pattern keeps only words of two or more characters and would drop "£", "!", "i", "u", "2" and "4".

Vocabulary growth. Both corpora are long-tailed (54% of Enron tokens also occur once). The 20% SMS dev split contains only 3,362 distinct tokens, fewer than the $n = 4{,}096$ originally proposed.

Feasibility. After the dev split and test set, SMS leaves $m_{\max} = 3{,}128$ and Enron 22,110. On SMS, $m_{\max}$ falls below $10n$ beyond $n \approx 300$. Where a crossover lies beyond $m_{\max}$ we can report only a lower bound.

5. Data Preprocessing and Feature Decisions

Decision	**Implementation**	**Rationale**
Deduplication	Drop exact repeats after lower-casing & whitespace collapse.	Prevents verbatim repeats from leaking across splits.
Stratified dev split	20% of each class held out for mutual information ranking.	Prevents vocabulary selection from accessing test data.
Stratified test/train	Paired draws matching pool rate; min 1 example per class.	Prevents near-single-class test sets & undefined priors at small m.
Tokenizer & Features	Custom pattern `[a-z0-9]+|[£$€!]`; Binary bag-of-words.	Retains high-signal digits/symbols; matches Ng & Jordan (2001).
Vocab Ranking	Top-n tokens by mutual information on dev split.	Prioritizes rare high-signal tokens over non-discriminative ones.

No hand-engineered features, stemming, or stop-word removals are applied, ensuring the feature space remains identical to the paper's theoretical framework.

6. Literature Review

This project relies entirely on Ng and Jordan (2001). The authors demonstrate that naive Bayes and logistic regression model the same joint distribution family. Naive Bayes fits $P(x, y)$ assuming conditional independence, converging in $O(\log n)$ examples, whereas logistic regression fits $P(y|x)$ directly, requiring $O(n)$ examples. While naive Bayes performs better at small training sizes, logistic regression achieves lower asymptotic error, creating a theoretical crossover point $m^*(n)$. This project tests these sample complexity bounds empirically under identical feature representations.

7. Initial Plan for Model Development and Experimentation

We will evaluate Multivariate Bernoulli naive Bayes with Laplace smoothing ($\alpha = 1$) alongside unregularized/default Logistic Regression (L_2 penalty, default solver). Pipeline sweeps will evaluate a hand-picked grid of sample sizes m using Monte Carlo repeated train/test splits. Paired t-tests across repeated runs will evaluate significance at each m point. Both 0–1 test error and log-loss will be recorded for every fit. Multinomial naive Bayes will be run alongside as a baseline, and the proposal's target of 200 or more repeated splits per corpus sets the scale of the sweep.

8. Project Roadmap and Progress

Stage 0 (Brainstorm/Feasibility) and Stage 0.5 (EDA/Preprocessing) are complete. Stage 1 (Data Pipeline) is currently in progress. Stages 2–9 (Model implementation, sweeps, significance testing, and paper write-up) have not been started. The immediate milestone is establishing a single end-to-end split harness, with the proposal's first gate being an end-to-end run at $n = 64$. The later gates follow the proposal's order: all (n, m) runs logged, then results frozen after the significance tests, crossover estimates and scaling fits, then the paper and presentation.

9. Team-Member Roles and Contributions

Owner	Responsibility	Status
Kevin Liu	Data pipeline, corpus cleaning, feature matrices, split/training harness, experiment execution.	Corpus cleaning and EDA complete. Pipeline design in progress.
Andrew Ngo	Model implementation, statistical analysis, crossover estimation, scaling fits.	Not started; pending data pipeline completion.
Bryce Lee	Theoretical mapping, figure production, report drafting, presentation slides.	Literature review drafted; figures/write-up pending experiment results.

References

Ng, A. Y., & Jordan, M. I. (2001). On discriminative vs. generative classifiers: A comparison of logistic regression and naive Bayes. *Advances in Neural Information Processing Systems 14*, 841–848.

Almeida, T. A., Gómez Hidalgo, J. M., & Yamakami, A. (2012). SMS Spam Collection. UCI Machine Learning Repository.

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